DAG

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Souza Ranch

Physical address of dairy:

3842 Villa Manucha Rd.	Newman	Stanislaus	95360
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 11/01/1981

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X049-X024-X002-XXXX

B. OPERATORS

Souza, Anthony J

Operator name: Souza, Anthony J

Telephone no.: (209) 837-4792 Landline, (209) 456-6875 Cellular

3842 Villamanucha RD	Newman	CA	95360
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Souza, Anthony J

Legal owner name: Souza, Anthony J

Telephone no.: (209) 837-4792 Landline, (209) 456-6875 Cellular

3842 Villamanucha RD	Newman	CA	95360
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	150	50	0	0
Number under roof	500	75	0	0	0	0
Maximum number	500	75	150	50	0	0
Average number	500	75	150	50	0	0
Avg live weight (lbs)	1,100	1,150	750	550		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 50 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 14,544.04 *tons per reporting period*

Total nitrogen from manure: 185,012.95 *lbs per reporting period*

After ammonia losses (30% loss applied): 129,509.07 *lbs per reporting period*

Total phosphorus from manure: 30,586.22 *lbs per reporting period*

Total potassium from manure: 89,184.61 *lbs per reporting period*

Total salt from manure: 252,671.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 12,540,000 *gallons*

Total nitrogen generated: 45,232.24 *lbs*

Total phosphorus generated: 5,774.97 *lbs*

Total potassium generated: 64,780.82 *lbs*

Total salt generated: 506,047.48 *lbs*

	12,540,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	12,540,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
CCID	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
08/02/2023	Solid commercial fertilizer *UN 32*	4.15 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	2,653.34	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	2,653.34	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Dairy	78	78	2	process wastewater	X049-X024-X002-XXXX
Fantazia	13	13	2	manure	X049-X024-X007-XXXX X049-X024-X009-XXXX
Serpa	39	39	2	manure	X049-X026-X014-XXXX X049-X026-X015-XXXX
Silveira	30	30	2	process wastewater	X049-X026-X005-XXXX
Stuhr	31	31	2	manure	X049-X031-X005-XXXX
Totals for areas that were used for application	191	191	10		
Totals for areas that were not used for application					
Land application area totals	191	191	10		

B. CROPS AND HARVESTS

Dairy

Field name: Dairy

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 78 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/30/2023	1,191.00 *ton*	As-is		69.0	3,410.00	800.00	5,300.00		9.16

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.27	104.14	24.43	161.85	867.17

Dairy

06/30/2023: Corn, silage

Crop: Corn, silage — Acres planted: 78 — Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	2,033.00 *ton*	As-is		74.7	4,020.00	1,340.00	11,200.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.06	209.56	69.85	583.84	1,345.22

Fantazia

Field name: Fantazia

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough — Acres planted: 13 — Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/31/2023	182.00 *ton*	As-is		60.8	4,080.00	970.00	6,670.00		9.02

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.00	114.24	27.16	186.76	990.04

06/30/2023: Corn, silage

Crop: Corn, silage — Acres planted: 13 — Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	325.00 *ton*	As-is		73.4	4,390.00	1,270.00	10,700.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.00	219.50	63.50	535.00	1,502.90

Serpa

Field name: Serpa

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough — Acres planted: 39 — Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/31/2023	591.00 *ton*	As-is		62.0	3,420.00	680.00	4,760.00		7.71

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.15	103.65	20.61	144.26	887.95

07/01/2023: Corn, silage

Crop: Corn, silage — Acres planted: 39 — Plant date: 07/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	994.00 *ton*	As-is		74.9	3,920.00	1,490.00	12,900.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.49	199.82	75.95	657.57	1,343.43

Silveira

Field name: Silveira

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough — Acres planted: 30 — Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/30/2023	504.00 *ton*	As-is		68.0	3,010.00	750.00	5,230.00		8.74

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.80	101.14	25.20	175.73	939.72

Silveira

06/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	811.00 *ton*	As-is		74.2	4,210.00	1,430.00	12,700.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	27.03	227.62	77.32	686.65	1,757.60

Stuhr

Field name: Stuhr

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 31 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/30/2023	542.00 *ton*	As-is		70.2	2,950.00	680.00	4,710.00		9.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	17.48	103.15	23.78	164.70	943.05

07/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 31 Plant date: 07/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	906.00 *ton*	As-is		75.3	3,750.00	1,320.00	11,000.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	29.23	219.19	77.16	642.97	1,559.26

NUTRIENT BUDGET

A. LAND APPLICATIONS

Dairy - 11/14/2022: Oats, silage-soft dough

Field name: Dairy

Crop: Oats, silage-soft dough

Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.84	7.52	56.16	450.63	1,080,000.00 *gal*
Application event totals		50.84	7.52	56.16	450.63	

02/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.19	6.69	49.92	400.56	960,000.00 *gal*
Application event totals		45.19	6.69	49.92	400.56	

04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.21	4.90	65.42	453.97	960,000.00 *gal*
Application event totals		38.21	4.90	65.42	453.97	

Dairy - 06/30/2023: Corn, silage

Field name: Dairy

Crop: Corn, silage

Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	85.97	11.02	147.21	1,021.43	2,160,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	225.66	13,608,000.00 *gal*
Application event totals		85.97	11.02	147.21	1,247.09	

Dairy - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	218.50	13,176,000.00 *gal*
Application event totals		0.00	0.00	0.00	218.50	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.78	11.48	138.66	1,162.65	1,920,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	216.71	13,068,000.00 *gal*
Application event totals		97.78	11.48	138.66	1,379.36	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	200.59	12,096,000.00 *gal*
Application event totals		0.00	0.00	0.00	200.59	

09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.67	10.77	129.99	1,089.99	1,800,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	209.54	12,636,000.00 *gal*
Application event totals		91.67	10.77	129.99	1,299.53	

09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	205.96	12,420,000.00 *gal*
Application event totals		0.00	0.00	0.00	205.96	

Fantazia - 11/14/2022: Oats, silage-soft dough

Field name: Fantazia

Crop: Oats, silage-soft dough

Plant date: 11/14/2022

Fantazia - 11/14/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	132.75	39.28	214.44	2,430.35	50.00 *ton*
Application event totals		132.75	39.28	214.44	2,430.35	

Fantazia - 06/30/2023: Corn, silage

Field name: Fantazia

Crop: Corn, silage

Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	263.38	98.51	450.86	0.00	120.00 *ton*
Application event totals		263.38	98.51	450.86	0.00	

06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	257.90	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	257.90	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
CCID	Surface water	0.00	0.00	0.00	214.92	2,160,000.00 *gal*
Application event totals		32.00	0.00	0.00	214.92	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	225.66	2,268,000.00 *gal*
Application event totals		0.00	0.00	0.00	225.66	

Fantazia - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	214.92	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	214.92	

09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	204.17	2,052,000.00 *gal*
Application event totals		0.00	0.00	0.00	204.17	

09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	193.42	1,944,000.00 *gal*
Application event totals		0.00	0.00	0.00	193.42	

Serpa - 11/15/2022: Oats, silage-soft dough

Field name: Serpa

Crop: Oats, silage-soft dough Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	132.75	39.28	214.44	2,430.35	150.00 *ton*
Application event totals		132.75	39.28	214.44	2,430.35	

Serpa - 07/01/2023: Corn, silage

Field name: Serpa

Crop: Corn, silage Plant date: 07/01/2023

Serpa - 07/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	234.11	87.56	400.77	0.00	320.00 *ton*
Application event totals		234.11	87.56	400.77	0.00	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	218.50	6,588,000.00 *gal*
Application event totals		0.00	0.00	0.00	218.50	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
CCID	Surface water	0.00	0.00	0.00	214.92	6,480,000.00 *gal*
Application event totals		32.00	0.00	0.00	214.92	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	207.75	6,264,000.00 *gal*
Application event totals		0.00	0.00	0.00	207.75	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	200.59	6,048,000.00 *gal*
Application event totals		0.00	0.00	0.00	200.59	

09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	193.42	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	193.42	

Serpa - 07/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	189.84	5,724,000.00 *gal*
Application event totals		0.00	0.00	0.00	189.84	

Silveira - 11/14/2022: Oats, silage-soft dough

Field name: Silveira

Crop: Oats, silage-soft dough Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.75	8.69	64.89	520.73	480,000.00 *gal*
Application event totals		58.75	8.69	64.89	520.73	

02/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.39	5.98	44.61	358.00	330,000.00 *gal*
Application event totals		40.39	5.98	44.61	358.00	

04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.15	4.38	58.47	405.73	330,000.00 *gal*
Application event totals		34.15	4.38	58.47	405.73	

Silveira - 06/30/2023: Corn, silage

Field name: Silveira

Crop: Corn, silage Plant date: 06/30/2023

Silveira - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	86.92	11.15	148.84	1,032.78	840,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	246.80	5,724,000.00 *gal*
Application event totals		86.92	11.15	148.84	1,279.57	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	228.17	5,292,000.00 *gal*
Application event totals		0.00	0.00	0.00	228.17	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	111.22	13.06	157.72	1,322.52	840,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	223.51	5,184,000.00 *gal*
Application event totals		111.22	13.06	157.72	1,546.03	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	223.51	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	223.51	

09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	111.22	13.06	157.72	1,322.52	840,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	214.20	4,968,000.00 *gal*
Application event totals		111.22	13.06	157.72	1,536.72	

Stuhr - 07/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	229.82	5,508,000.00 *gal*
Application event totals		0.00	0.00	0.00	229.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
CCID	Surface water	0.00	0.00	0.00	225.32	5,400,000.00 *gal*
Application event totals		32.00	0.00	0.00	225.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	225.32	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	225.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	216.30	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	216.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	216.30	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	216.30	

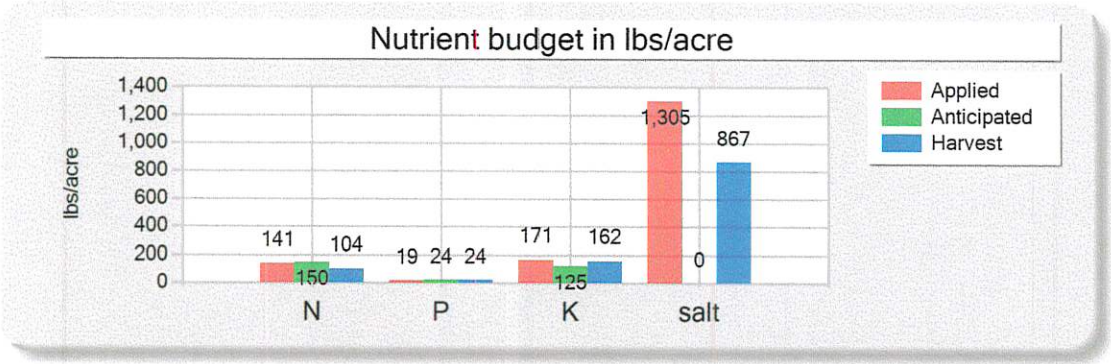
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	207.29	4,968,000.00 *gal*
Application event totals		0.00	0.00	0.00	207.29	

B. NUTRIENT BUDGET

Dairy - 11/14/2022: Oats, silage-soft dough

Field name: Dairy Crop: Oats, silage-soft dough Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	134.24	19.11	171.50	1,305.16
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	141.24	19.11	171.50	1,305.16
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	104.14	24.43	161.85	867.17
Nutrient balance	37.10	-5.32	9.64	437.99
Applied to removed ratio	1.36	0.78	1.06	1.51

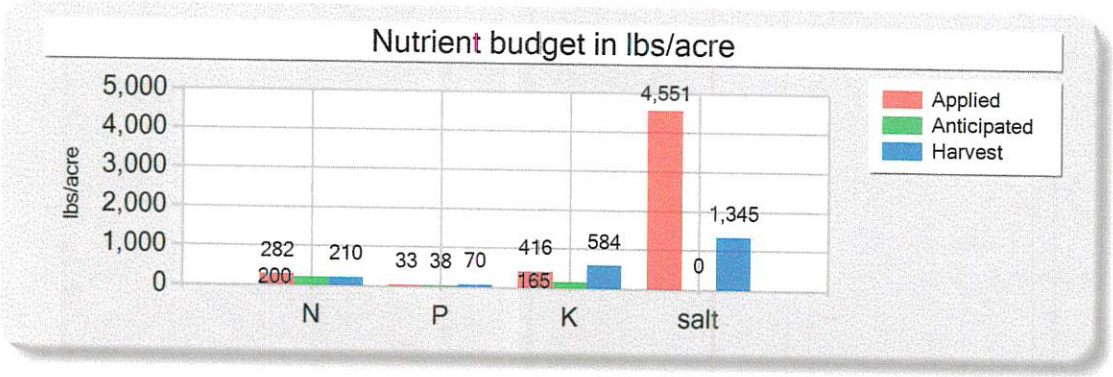
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
3,000,000.00 *gallons*
110.48 *acre-inches*
1.42 *inches/acre*

Total harvests for the crop
1 *harvests*

Dairy - 06/30/2023: Corn, silage

Field name: Dairy Crop: Corn, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	275.41	33.27	415.85	3,274.06
Fresh water	0.00	0.00	0.00	1,276.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.41	33.27	415.85	4,551.02
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	209.56	69.85	583.84	1,345.22
Nutrient balance	72.86	-36.58	-167.99	3,205.80
Applied to removed ratio	1.35	0.48	0.71	3.38

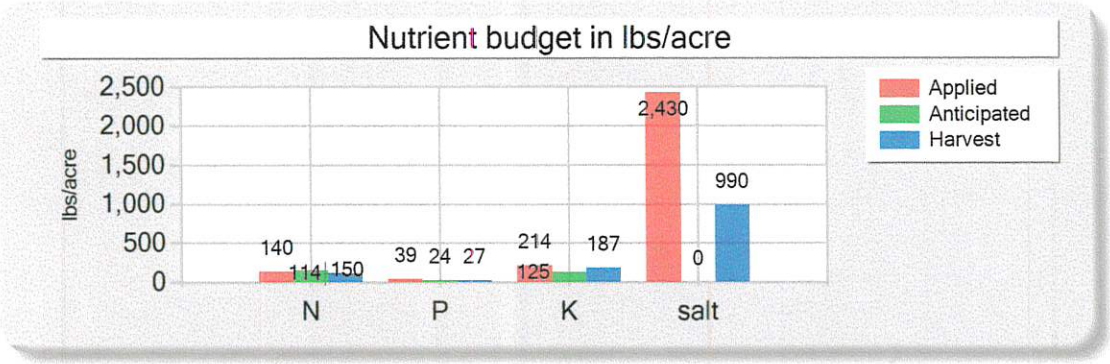
Fresh water applied
77,004,000.00 *gallons*
2,835.80 *acre-inches*
36.36 *inches/acre*

Process wastewater applied
5,880,000.00 *gallons*
216.54 *acre-inches*
2.78 *inches/acre*

Total harvests for the crop
1 *harvests*

Fantazia - 11/14/2022: Oats, silage-soft dough

Field name: Fantazia Crop: Oats, silage-soft dough Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	132.75	39.28	214.44	2,430.35
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	139.75	39.28	214.44	2,430.35
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	114.24	27.16	186.76	990.04
Nutrient balance	25.51	12.12	27.68	1,440.31
Applied to removed ratio	1.22	1.45	1.15	2.45

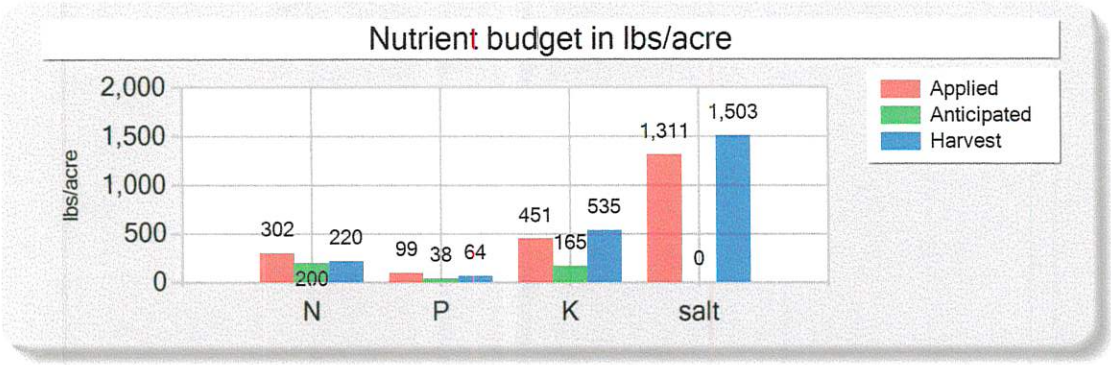
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Fantazia - 06/30/2023: Corn, silage

Field name: Fantazia | Crop: Corn, silage | Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	263.38	98.51	450.86	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,310.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.38	98.51	450.86	1,310.99
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	219.50	63.50	535.00	1,502.90
Nutrient balance	82.88	35.01	-84.14	-191.91
Applied to removed ratio	1.38	1.55	0.84	0.87

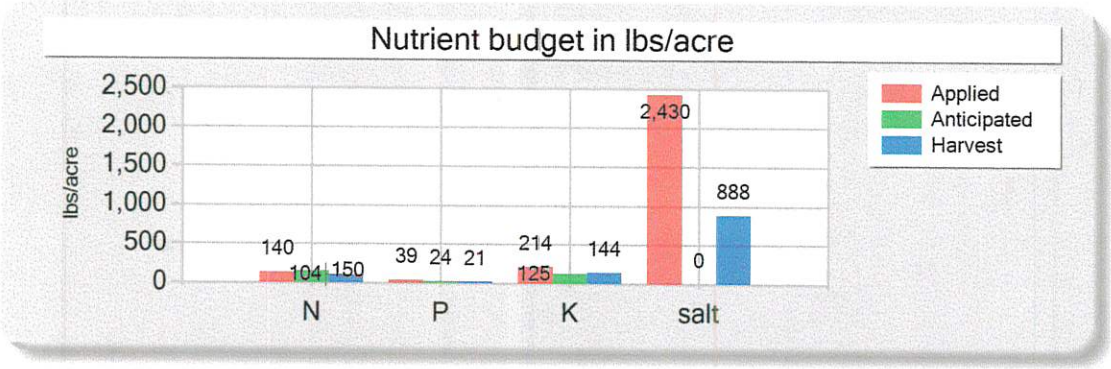
Fresh water applied
13,176,000.00 *gallons*
485.23 *acre-inches*
37.33 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Serpa - 11/15/2022: Oats, silage-soft dough

Field name: Serpa Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	132.75	39.28	214.44	2,430.35
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	139.75	39.28	214.44	2,430.35
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	103.65	20.61	144.26	887.95
Nutrient balance	36.10	18.67	70.18	1,542.39
Applied to removed ratio	1.35	1.91	1.49	2.74

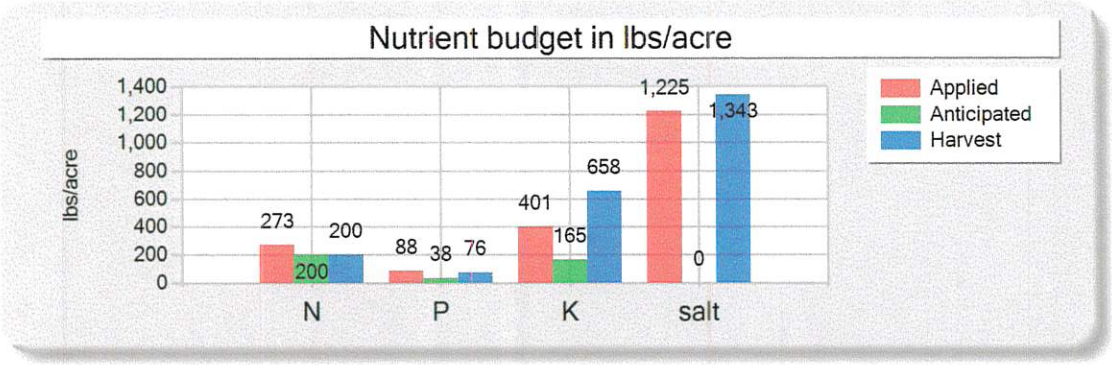
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Serpa - 07/01/2023: Corn, silage

Field name: Serpa | Crop: Corn, silage | Plant date: 07/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	234.11	87.56	400.77	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,225.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	273.11	87.56	400.77	1,225.02
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	199.82	75.95	657.57	1,343.43
Nutrient balance	73.29	11.61	-256.80	-118.41
Applied to removed ratio	1.37	1.15	0.61	0.91

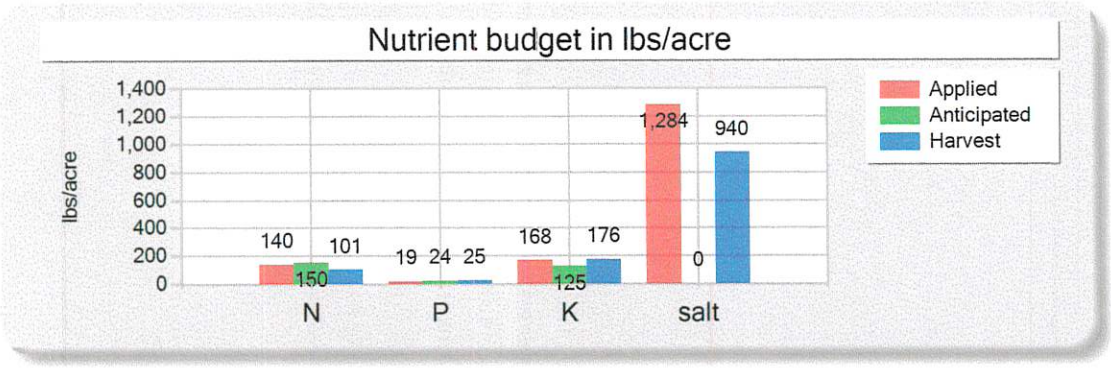
Fresh water applied
36,936,000.00 *gallons*
1,360.23 *acre-inches*
34.88 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Silveira - 11/14/2022: Oats, silage-soft dough

Field name: Silveira | Crop: Oats, silage-soft dough | Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	133.29	19.05	167.98	1,284.46
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	140.29	19.05	167.98	1,284.46
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	101.14	25.20	175.73	939.72
Nutrient balance	39.15	-6.15	-7.75	344.74
Applied to removed ratio	1.39	0.76	0.96	1.37

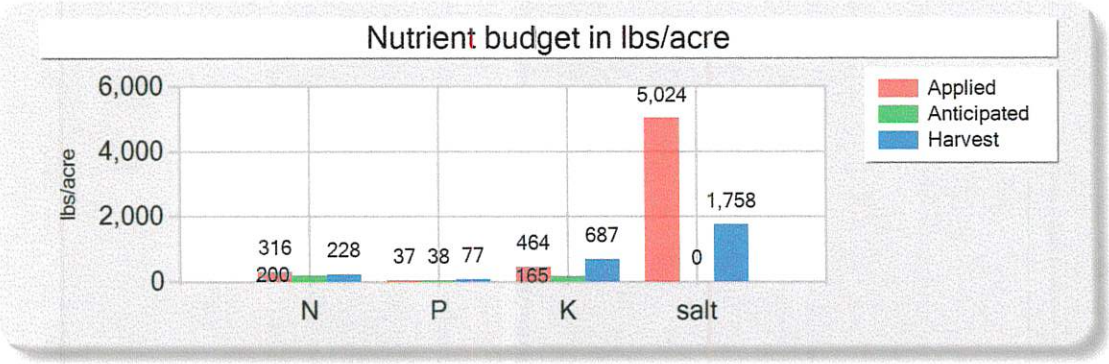
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,140,000.00 *gallons*
41.98 *acre-inches*
1.40 *inches/acre*

Total harvests for the crop
1 *harvests*

Silveira - 06/30/2023: Corn, silage

Field name: Silveira | Crop: Corn, silage | Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	309.37	37.27	464.28	3,677.81
Fresh water	0.00	0.00	0.00	1,345.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	316.37	37.27	464.28	5,023.54
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	227.62	77.32	686.65	1,757.60
Nutrient balance	88.75	-40.05	-222.36	3,265.94
Applied to removed ratio	1.39	0.48	0.68	2.86

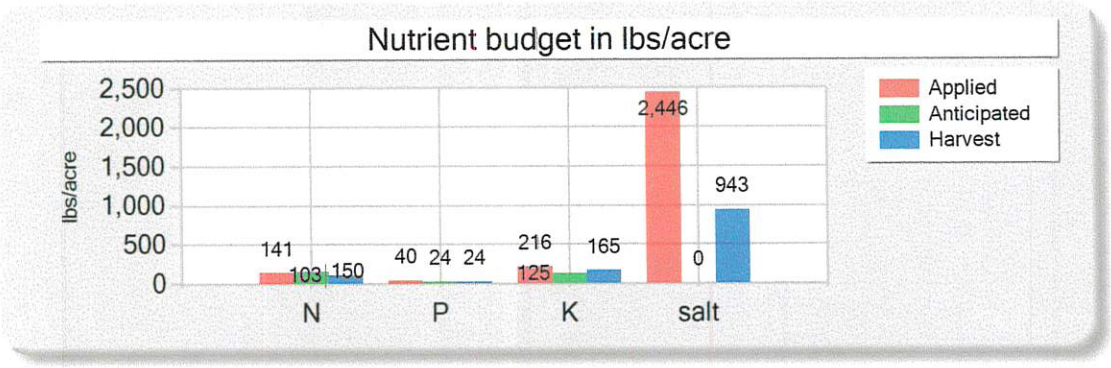
Fresh water applied
31,212,000.00 *gallons*
1,149.43 *acre-inches*
38.31 *inches/acre*

Process wastewater applied
2,520,000.00 *gallons*
92.80 *acre-inches*
3.09 *inches/acre*

Total harvests for the crop
1 *harvests*

Stuhr - 11/15/2022: Oats, silage-soft dough

Field name: Stuhr | Crop: Oats, silage-soft dough | Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	133.61	39.53	215.83	2,446.03
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	140.61	39.53	215.83	2,446.03
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	103.15	23.78	164.70	943.05
Nutrient balance	37.45	15.76	51.13	1,502.98
Applied to removed ratio	1.36	1.66	1.31	2.59

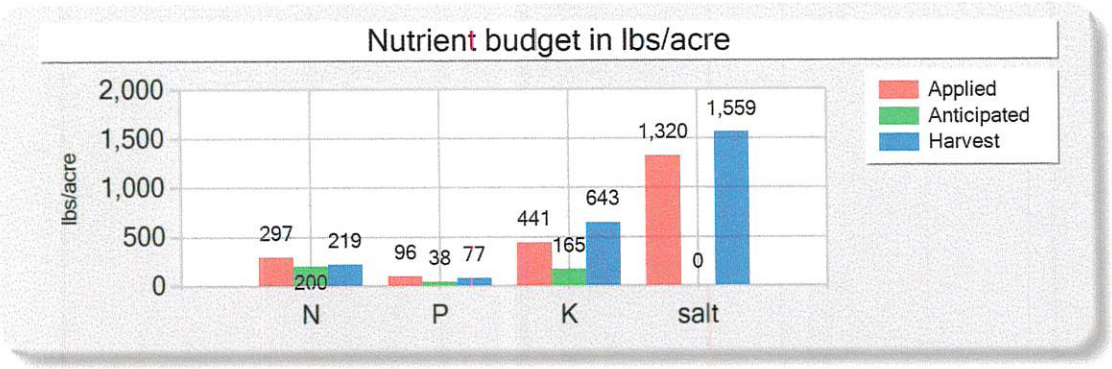
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Stuhr - 07/01/2023: Corn, silage

Field name: Stuhr | Crop: Corn, silage | Plant date: 07/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	257.71	96.39	441.17	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,320.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	296.71	96.39	441.17	1,320.35
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	219.19	77.16	642.97	1,559.26
Nutrient balance	77.52	19.23	-201.80	-238.91
Applied to removed ratio	1.35	1.25	0.69	0.85

Fresh water applied
31,644,000.00 *gallons*
1,165.34 *acre-inches*
37.59 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure

Sample and source description: Manure

Sample date: 06/27/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 19.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	6,620.00	30,300.00							
DL	100.00	100.00	100.00							

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/20/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,500.00	5,770.00	31,500.00	21,400.00	8,490.00	4,520.00	6,900.00	1,480.00		35.70
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

Lagoon

Sample and source description: Lagoon

Sample date: 06/27/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	372.00	249.00	0.00	0.00	47.70	637.00								6,550.00	4,420
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Lagoon 3QWW

Sample and source description: Lagoon 3QWW

Sample date: 10/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	476.00	308.00	0.00	0.00	55.90	675.00								7,160.00	5,660
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Lagoon

Sample and source description: Lagoon

Sample date: 12/30/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	440.00	193.00	0.00	0.00	65.10	486.00								4,920.00	3,900
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

CCID

CCID

Sample description: CCID

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								228.00	155
DL	1.00	0.50	0.10								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Fantazia - 06/30/2023: Corn, silage

Fantazia

Sample and source description: Fantazia

Sample date: 10/20/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,390.00	1,270.00	10,700.00		11.30
DL	100.00	100.00	100.00		0.01

Serpa - 11/15/2022: Oats, silage-soft dough

Serpa

Sample and source description: Serpa

Sample date: 06/27/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,420.00	680.00	4,760.00		7.71
DL	100.00	100.00	100.00		0.01

Serpa - 07/01/2023: Corn, silage

Serpa

Sample and source description: Serpa

Sample date: 10/20/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,920.00	1,490.00	12,900.00		10.50
DL	100.00	100.00	100.00		0.01

Silveira - 11/14/2022: Oats, silage-soft dough

Silveira

Sample and source description: Silveira

Sample date: 06/27/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,010.00	750.00	5,230.00		8.74
DL	100.00	100.00	100.00		0.01

Silveira - 06/30/2023: Corn, silage

Silveira

Sample and source description: Silveira

Sample date: 10/20/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,210.00	1,430.00	12,700.00		12.60
DL	100.00	100.00	100.00		0.01

Stuhr - 11/15/2022: Oats, silage-soft dough

Stuhr

Sample and source description: Stuhr

Sample date: 06/27/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,950.00	680.00	4,710.00		9.05
DL	100.00	100.00	100.00		0.01

Stuhr - 07/01/2023: Corn, silage

Stuhr

Sample and source description: Stuhr

Sample date: 10/20/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,750.00	1,320.00	11,000.00		10.80
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

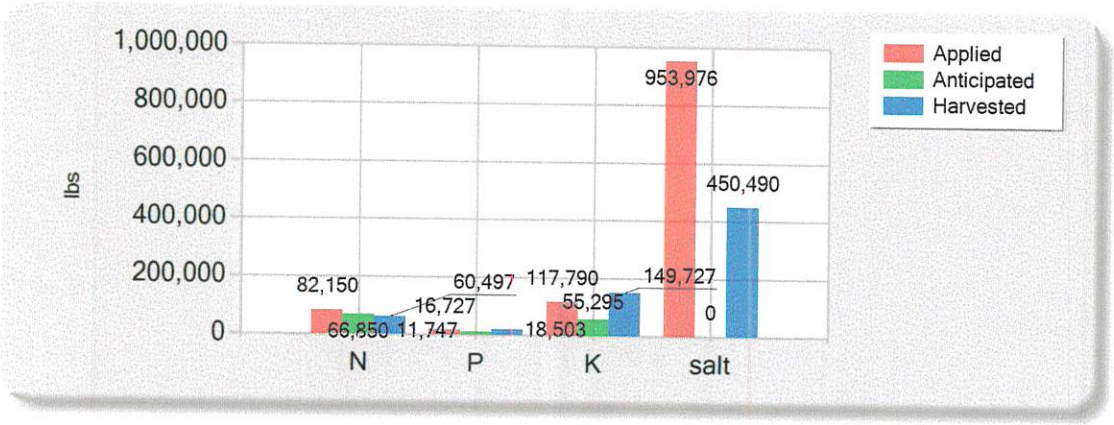
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

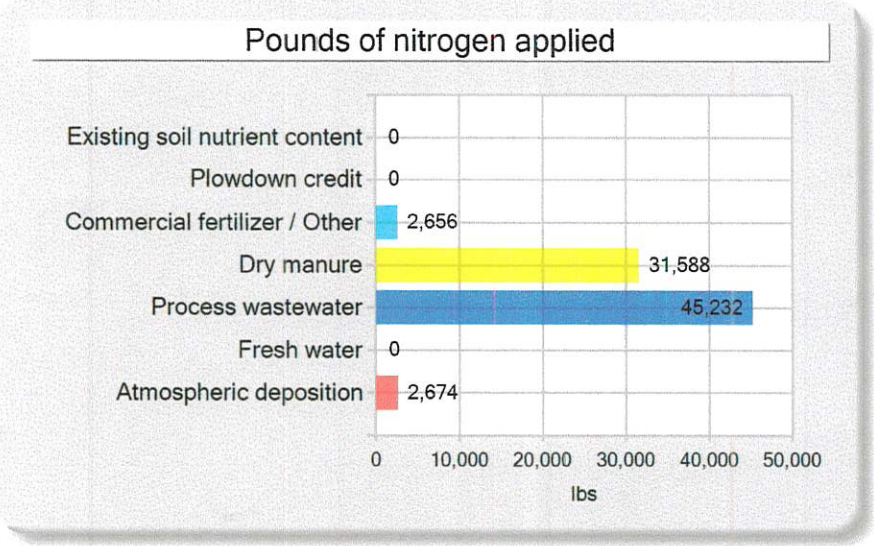
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

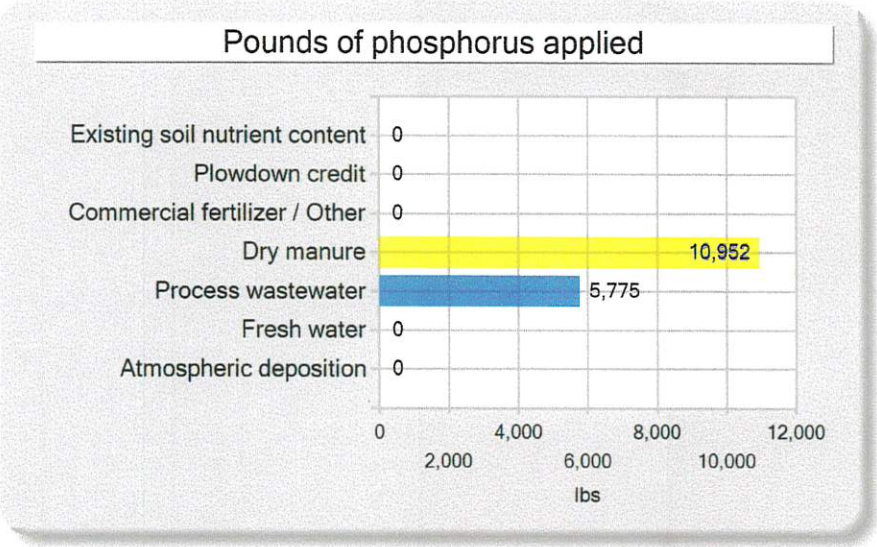
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	2,656.00	0.00	0.00	0.00
Dry manure	31,588.13	10,951.56	53,008.99	202,204.80
Process wastewater	45,232.24	5,774.97	64,780.82	506,047.48
Fresh water	0.00	0.00	0.00	245,724.03
Atmospheric deposition	2,674.00	0.00	0.00	0.00
Total nutrients applied	82,150.36	16,726.54	117,789.81	953,976.31
Anticipated crop nutrient removal	66,850.00	11,746.50	55,294.50	0.00
Actual crop nutrient removal	60,497.46	18,502.92	149,727.08	450,489.63
Nutrient balance	21,652.90	-1,776.38	-31,937.27	503,486.68
Applied to removed ratio	1.36	0.90	0.79	2.12

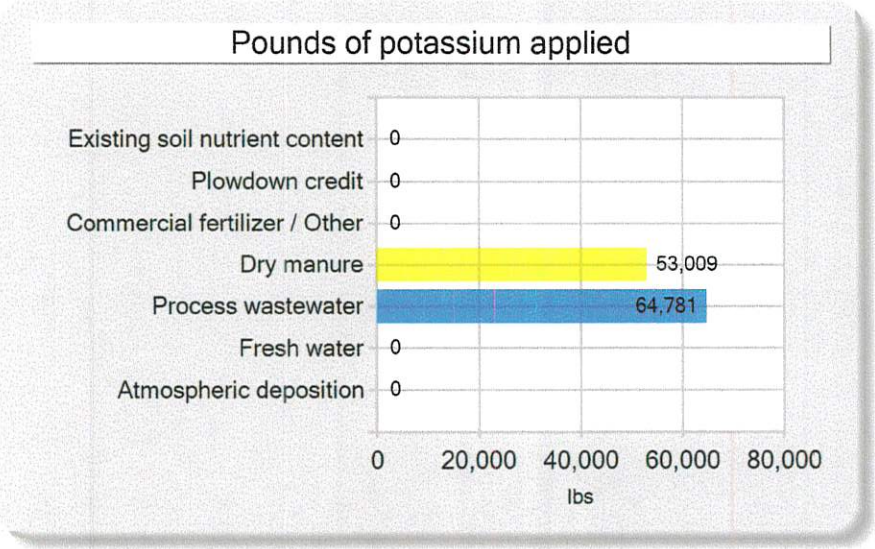
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

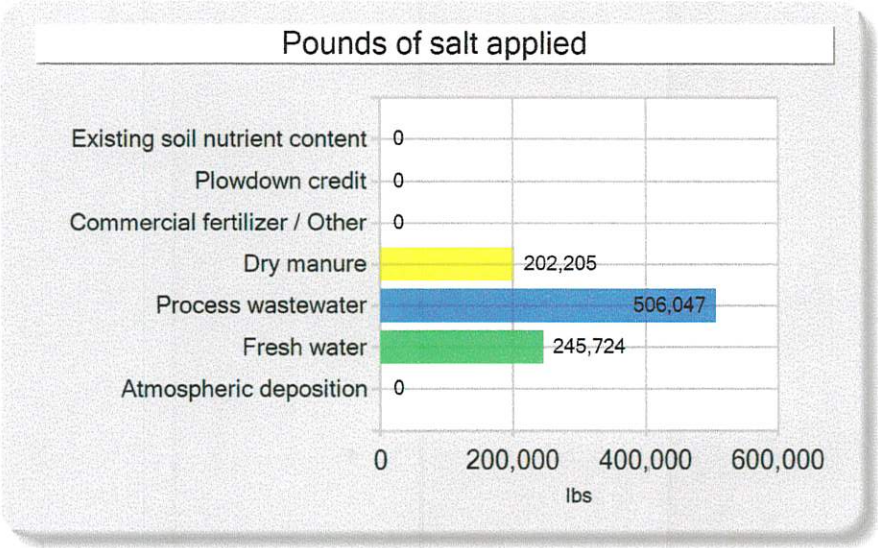


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Anthony J Souza	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-26-2024	6-26-2024
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted.

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed.

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.